Ai-assisted adaptive questioning in survey research

Aku Antero Paloheimo
Aalto University
Espoo, Finland
aku.paloheimo@aalto.fi

Oona Anniina Tujula
Aalto University
Espoo, Finland
oona.tujula@aalto.fi

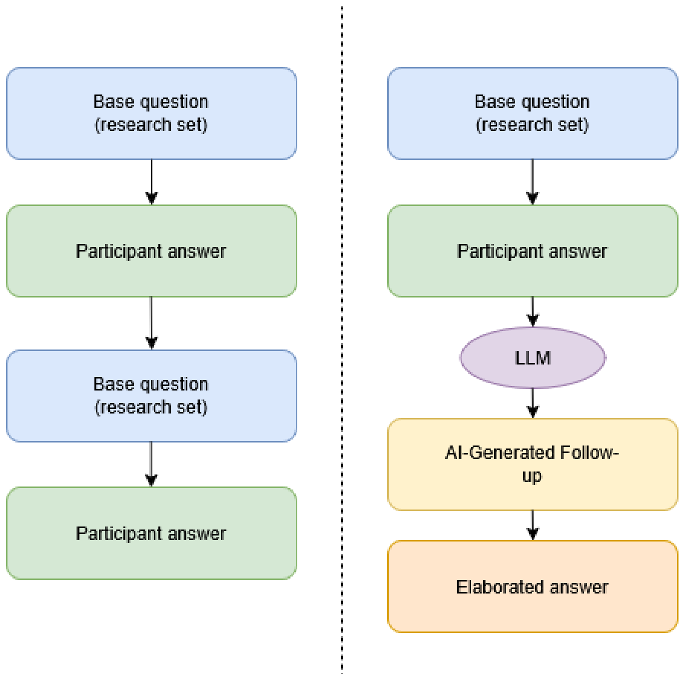

Figure 1: Comparison between a traditional fixed survey flow and the AI-assisted adaptive survey flow. In the AI-assisted condition, the language model receives the base question and the participant's answer, and generates a follow-up question to encourage elaboration.

Abstract

Surveys are scalable and easy to administer, but their fixed nature can lack depth. Semi-structured interviews produce more nuanced results and offer opportunities to gain a better understanding of the participant's answers, however they are often very demanding to conduct for researchers. Recent progress in large language models (LLMs) suggests new ways to rethink how survey-based research is designed and analyzed in Human–Computer Interaction (HCI).

This study examines AI-assisted adaptive questioning in online self-administered surveys, where foundation models create follow-up questions in real time based on each participants' answers. In addition to examining this methodological problem, the study presents a lightweight adaptive survey artefact that operationalizes LLM-generated follow-up questioning in a controlled research setting. The contribution of this work is twofold: (1) a system contribution in the form of an AI-assisted adaptive survey platform, and (2) a methodological contribution that reflects on the epistemic and practical implications of delegating follow-up questioning to a large language model and then showing those generated questions to participants.

In an empirical study, we compare responses from AI-supported adaptive surveys and conventional fixed surveys based on the same five open-ended base questions in order to examine differences in qualitative depth, clarity, and relevance.

The findings suggest that AI-assisted surveys can produce more detailed responses, but they also create new challenges in comparing results and maintaining methodological control. These findings add to what we know about when LLM-enabled surveys work well and where they fall short as a supporting tool in early-stage HCI research. They also offer practical advice for designing adaptive surveys that keep rich qualitative input while still meeting standards for methodological rigor.

Conference'17, Washington, DC, USA

ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/XXXXXXX.XXXXXXX

Keywords

HCI, LLMs, Survey research, AI

ACM Reference Format:
Aku Antero Paloheimo and Oona Anniina Tujula. 2018. Ai-assisted adaptive questioning in survey research. In . ACM, New York, NY, USA, 7 pages. https://doi.org/XXXXXXX.XXXXXXX

1 Introduction

Due to their low cost and scalability, surveys are one of the most commonly used data collection methods in Human Computer Interaction (HCI) research [4][9]. However, traditional surveys are limited by their static nature [8]. Quantitative questions, such as multiple-choice or Likert scales, provide structured data that is easily analyzed but limit the surveyees ability to fully express their perspectives [4][8]. On the other hand, open-ended qualitative questions allow for more detailed answers, but the questions are often very broad and difficult to answer [4][8]. Additionally the resulting unstructured data is labor-intensive to process and hard to generalise for studies [4]. Therefore surveys are naturally limited by the value they can bring, where respondents can only answer predefined questions, and researchers have no opportunity to ask clarifying follow-up questions based on individual responses.

Recent advances in large language models have made conversational artificial intelligence (AI) tools more accessible and affordable to use in research [2]. This creates an opportunity to reconsider traditional survey methodologies. LLMs could come up with follow-up questions that encourage more elaborated answers that could cover some of the things that may get overlooked while making the survey. Investigating AI-supported surveys is therefore of significance to both academic HCI research and industry practitioners seeking richer insights from survey-based studies, particularly in contexts where conducting large numbers of interviews is impractical.

Past research has examined how LLMs can be used in surveys and qualitative analysis in HCI, but it remains unclear whether AI-generated adaptive follow-up questions can close the methodological gap between fixed surveys and semi-structured interviews [2, 7, 8]. It is still uncertain whether this kind of adaptivity can improve the depth, clarity, and relevance of responses without weakening the comparability and rigor expected in survey research [2, 5]. However, there is a clear need to study how LLMs can be used ethically and more effectively. Moreover, the extensive use of LLMs in survey research has been limited. [5].

To this end, this research investigates the potential use of LLMs in survey-based research through an AI-assisted survey platform, to ease the gathering and processing of qualitative data in HCI research. The research aims to provide empirical insight into the effectiveness of AI assisted surveys in capturing more qualitative data compared to conventional surveys. Second, it offers methodological guidance for HCI researchers by outlining design considerations, benefits, and limitations of incorporating AI into research. The research also aims to examine how respondents experience AI-driven follow-up questions compared to fixed survey questions to identify practical challenges and ethical considerations with using LLMs in survey-based research. By doing so, the research positions AI not as a replacement for qualitative methods such as interviews, but as a complementary tool that can enhance survey research.

Additionally the scope of the research is limited to early stage and exploratory research. These objectives are addressed through the following research question: Can AI-supported adaptive surveys produce data of greater depth, clarity, and relevance than conventional fixed surveys?

With the survey system, we evaluate the effectiveness of AI-assisted surveys through a comparative field study by conducting a conventional fixed survey and an AI-assisted adaptive survey using the same five base questions. The conventional survey was completed by four participants, and the AI-assisted survey was completed by four participants. Participants in both conditions represented similar demographic backgrounds. A semi-structured interview condition was considered as an alternative data collection method, but it was not included because the added condition did not fit in the time constraint the reserachers had.

In summary, we contribute:

(1) Empirical insight into the effectiveness of AI assisted surveys in capturing more qualitative data compared to conventional surveys.
(2) Methodological guidance for HCI researchers by outlining design considerations, benefits, and limitations of incorporating AI into research.
(3) Examine how respondents experience AI-driven follow-up questions compared to fixed survey questions to identify practical challenges and ethical considerations with using LLMs in survey-based research.

2 Related Work

2.1 Surveys in HCI research

In HCI research, surveys are widely used, as they are very easy to administer and do not require as extensive planning as some other forms of qualitative research, such as interviews [8]. Typically, a survey will consist of open-ended questions, closed-ended questions, or both. Closed-ended questions are generally easier to answer because they have a limited set of possible responses. However, they require careful design to ensure that respondents have the necessary knowledge to answer and interpret the question consistently [8].

On the contrary, open ended questions typically try to gather more qualitative data by asking broader questions. The problem often with this is that the researcher has to be very explicit in how they form their questions, since many respondents answer in very broad terms [8]. Additionally, broad question are more labor-intensive for both respondents and researchers. Open-ended questions can provide insight into differences in data quality between closed and open questions, especially when identical questions are used for comparison [8]. There is a need to see how open-ended and close-ended questions compare in Web questionnaires and another survey formats [8]. Despite these practical advantages, traditional surveys remain limited in their ability to support contingent follow-up questioning based on individual responses.

2.2 Large Language Models in HCI Research

It is said that easily accessible LLMs will change the way HCI researchers work with data [2]. LLMs have already been used to perform a number of tasks for researchers with differing results, such as qualitative coding, thematic analysis, and even mediating interviews or simulating user data [2]. However, there is a need to explore how to critically and ethically use LLM methods in HCI research [2]. Furthermore, the comparability of questions asked by LLMs has not been extensively studied.

In another workshop study conducted with HCI researchers described as "LLM curious", similar results were observed [7]. LLMs are already making a clear impact in HCI research and are expected to be used more widely in the future [7]. However, there is a clear need for standards and contextual norms that guide the ethical and valid use of LLMs in research [7].

2.3 LLMs in Survey Contexts

In a recent study, researchers built a survey platform that used LLMs to generate dynamic follow-up questions [5]. Their findings indicated that the platform provided valuable, quantifiable qualitative data and deeper insights for survey stakeholders, while also increasing engagement and fostering a sense of community. Future work could explore varied survey formats to address a broader range of researcher needs [5]. As the platform allowed only a single beginning question from which follow-up questions were generated, the authors suggested that future research could investigate surveys with multiple predetermined base questions [5].

The integration of AI into HCI systems introduces a range of ethical challenges that warrant careful consideration, particularly in research contexts. AI-assisted HCI systems have concerns including privacy, algorithmic bias, transparency, user autonomy, and accountability [1]. These concerns are especially pertinent when LLMs are deployed in data collection tools such as adaptive surveys, where participants may be unaware of the degree to which their responses are being shaped by automated probing.

Past research has primarily focused on how different qualitative interview and survey methods are perceived and how valid the resulting data are. LLMs have also been studied in this context. However, there is a clear need to study how LLMs can be used ethically and more effectively. Moreover, the extensive use of LLMs in survey research has been limited. This study aims to fill this gap in the literature.

3 Method

The study was conducted as a comparative study in which the researchers built an adaptive AI survey prototype and tested it against a conventional fixed survey. Both conditions used the same five base questions to ensure that differences in the data were not caused by the initial questions, but by the presence or absence of AI-generated follow-up questions.

The five questions that were used in the surveys were the following:

(1) Describe a hobby or activity that you do regularly, and what it typically involves?
(2) What motivates you to do this activity, and what does it mean to you personally?
(3) Can you walk me through a recent occasion when you engaged in this activity, including what you did and how you experienced it?

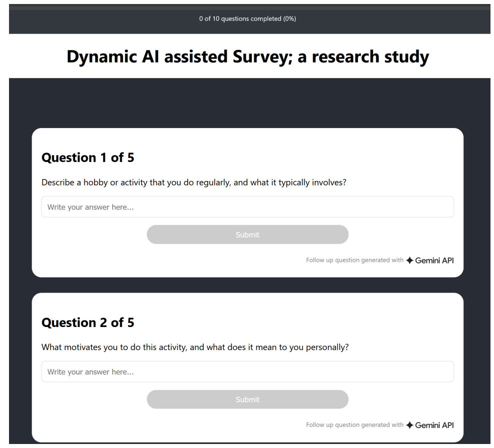

Figure 2: Screenshot of the prototype system used for collecting data in the AI-assisted survey condition

(4) What challenges or difficulties, if any, have you encountered in this activity?
(5) How has your engagement with this activity changed over time?

3.1 AI-Assisted Survey System

This section describes the AI assisted survey system that was designed to be tested against the regular survey.

A lightweight web-based survey platform was developed for the AI-assisted condition shown in Figure 2. The system presented one base question at a time and sent participants’ textual responses to the Google Gemini 3.0 Flash API in order to generate adaptive follow-up questions in real time. The main versions at the time of the experiment (March 2025) were: 3.1 Pro, the “best performing” multimodal model; 3.0 Pro, the previous version of the gemini model; 3.0 Flash, a cheaper variant “balancing performance and cost” (ai.google.dev/gemini-api/docs/models/gemini). Gemini 3.0 Flash was chosen because the more expensive alternative may not be practical for research evaluations.

The adaptive mechanism was governed by a structured system prompt. The model was instructed to generate one or two concise follow-up questions that sought clarification, elaboration, or concrete examples based strictly on the participant’s preceding answer. The prompt explicitly directed the model to avoid introducing unrelated topics, to maintain a neutral and non-judgmental tone, and to refrain from probing into sensitive personal domains. These constraints were designed to approximate the conversational norms of semi-structured interviewing while maintaining topical consistency across participants.

After a participant submitted a response to a base question, the response was programmatically sent to the language model. The generated follow-up question(s) were then displayed within the survey interface, and participants were invited to respond before proceeding to the next base question. This system is visualised in Figure 3 No human moderation occurred during live sessions; all follow-up prompts were generated automatically by the system.

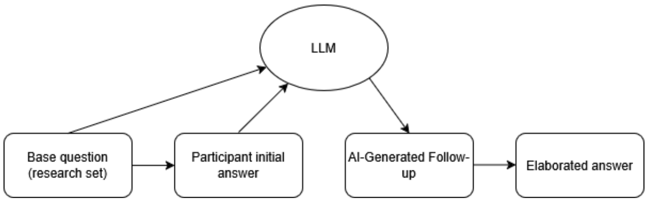

Figure 3: Visualisation of the adaptive interaction flow for a single question in the AI-assisted survey condition

Prior to data collection, the platform was piloted internally to verify technical stability, coherence of generated questions, and usability of the interaction flow. Minor adjustments were made to prompt phrasing to improve relevance and reduce redundancy in generated follow-ups.

In both conditions, participants first received information about the study and provided informed consent. In the conventional survey condition, participants completed the five base questions independently without adaptive follow-up questions. In the AI-assisted survey condition, participants completed the same five base questions through the online platform, where adaptive follow-up questions were generated automatically after each base response.

3.2 Study Design and Rationale

This study was designed as an exploratory methodological investigation examining whether AI-assisted adaptive surveys can produce richer and more detailed responses than conventional fixed surveys in early-stage HCI research. This work is primarily a methodological contribution. We use a prototype adaptive survey platform as a research instrument to examine the epistemic and practical implications of delegating follow-up questioning to a large language model.

The study adopts a comparative qualitative design with two conditions: a conventional fixed survey condition and an AI-assisted adaptive survey condition. The conventional fixed survey condition was selected as a baseline because it represents a common form of scalable survey-based data collection. This comparison allows the study to examine whether AI-generated follow-up questions can improve response depth while preserving the practical advantages of surveys.

A between-subjects design was employed, with four participants assigned to each condition. This choice was made to avoid carryover and learning effects that might arise if participants experienced both formats. The study is positioned as a feasibility exploration rather than a statistically generalizable comparison. Its goal is to examine patterns of qualitative richness and methodological implications rather than to produce inferential claims.

To ensure comparability across conditions, both survey formats were structured around the same five open-ended base questions. This design choice isolates the adaptive follow-up mechanism as the primary variable of interest, allowing differences in data quality to be interpreted in relation to how follow-up questions are generated and delivered.

Eight adult participants (N = 8) were recruited through convenience sampling [3]. Four participants completed the conventional fixed survey condition and four completed the AI-assisted survey condition. Participants represented varied demographic backgrounds in terms of age, gender, and educational experience. Participation was voluntary, and informed consent was obtained prior to data collection. No sensitive personal data were collected, and all responses were anonymized before analysis.

Both conditions were structured around five open-ended questions on the topic of hobbies. The topic was intentionally selected as non-sensitive and broadly accessible in order to reduce barriers to participation and minimize ethical risk, while still allowing space for elaboration and personal reflection. The questions were designed to invite descriptive detail and concrete examples, as well as to encourage participants to articulate motivations and meanings associated with their activities.

The wording of the five base questions was identical in both conditions. This consistency ensured that any observed differences in response depth or structure could be attributed primarily to the follow-up process rather than to variation in initial prompts.

3.3 Data Analysis

All textual data, including conventional survey responses and AI-assisted survey responses, were anonymized prior to analysis. The analysis followed an inductive qualitative coding approach. Open coding was first conducted to identify recurring concepts, descriptive elements, and forms of elaboration within responses. Codes were iteratively refined and subsequently clustered construct higher-level thematic groupings.

To examine comparative depth, the analysis attended to several qualitative indicators: the degree of descriptive elaboration, the presence of concrete examples, the articulation of motivations or reflections, and the overall coherence of responses. Rather than treating depth as a purely quantitative measure, it was conceptualized as the extent to which responses moved beyond surface-level statements toward contextualized, meaning-oriented accounts.

Both datasets were analyzed using the same coding framework to maintain analytic consistency. Given the exploratory scope and small sample size, the analysis emphasizes interpretive comparison and methodological reflection rather than statistical inference.

4 Results

This section presents findings from both conditions: four participants who completed the conventional survey and four who completed the AI-assisted adaptive survey, all responding to the same five open-ended base questions on the topic of hobbies.

4.1 Response Depth and Elaboration

Across the AI-assisted condition, participants consistently provided more detailed and contextualised responses in their follow-up answers than in their initial replies. A notable example is P1, whose initial description of gym training was brief and procedural. When prompted by the AI to articulate specific fitness goals, P1 produced a structured three-part account covering physical condition, health, and appearance, and subsequently revealed that these goals served as a means of maintaining mental health, an insight that did not surface in the initial answer. Similarly, P2 (boxing) expanded on

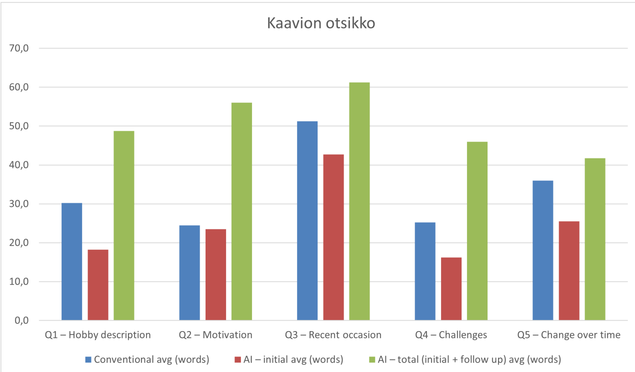

Figure 4: The figure shows the average word count answers to the conventional survey and in the AI assisted survey. From this we can see how respondents' word count did not follow a clear pattern but almost always the follow up questions made some significant contribution to the previous answer. Additionally the fatigue of the respondents' can be seen as the word count after question 3 reduces compared to the previous questions. Word count per response: initial answer vs. follow-up elaboration (n=4)

the personal significance of the activity when prompted, describing how managing physical hardship in training reframes everyday difficulties, providing a reflective and transferable account of meaning-making.

In the conventional survey condition, response depth varied considerably. P3 (pottery) and P4 (muay thai) produced rich and detailed initial answers without any prompting — P4 in particular provided a vivid account of a recent training session including situational detail, physical sensation, and emotional reflection. By contrast, P2 in the conventional condition (Python coding) provided minimal responses throughout, with initial answers averaging fewer than ten words per question. This variability suggests that conventional open-ended surveys are highly dependent on individual participant motivation and communication style, with no mechanism to compensate for brief or underspecified responses.

4.2 Relevance and Quality of AI-Generated Follow-Up Questions

The AI-generated follow-up questions were generally relevant and topically appropriate, engaging directly with the content of each participant's preceding answer rather than introducing unrelated themes. In most cases, the follow-ups sought either concrete examples, clarification of motivations, or elaboration on a specific statement, which are valid qualitative probing strategies used by human reasearchers also.

However, the findings also revealed important methodological limitations. Some generated follow-up questions implicitly constrained participant responses by embedding assumptions or predefined interpretive categories into the prompt itself. Rather than inviting open reflection, these prompts risked steering participants toward confirming the framing proposed by the model. This differs from ideal qualitative interviewing practice, where follow-up questions are generally phrased to minimize leading structures and preserve participant-driven meaning construction.

The issue was especially visible in cases where the AI generated multiple example interpretations within the same question. While such prompts may increase conversational fluency and reduce participant effort, they can also narrow the conceptual space of possible responses. As a result, participants may align their answers with the categories proposed by the system instead of articulating independently salient perspectives.

This highlights a broader methodological tension in AI-assisted qualitative data collection. More directive prompts may improve response length and apparent depth, yet simultaneously reduce the openness traditionally valued in qualitative inquiry. Designing adaptive survey systems therefore requires balancing conversational guidance with methodological neutrality.

4.3 Cases of Limited Elaboration

Not all AI-generated follow-ups produced meaningful elaboration. When P3 (podcasts) was asked to describe a specific instance where a podcast or ebook had changed their thinking, they responded "nothing specific comes to mind right now" indicating that the follow-up prompt was too demanding for the participant's level of reflective engagement at that moment. A human interviewer would likely have recognised this and moved on or reframed the question; the AI system, by contrast, had no mechanism for adapting to signs of disengagement mid-session.

A related pattern emerged when participants reported having no difficulties with their activity (P3, Q4: "I haven't really encountered any challenges"). The system generated a follow-up question regardless, producing a response that largely restated the original answer without adding substantive new information. This highlights a structural limitation of the current system: the AI applies follow-up questioning uniformly, without assessing whether the initial answer is already sufficient for research purposes.

4.4 Participant Engagement and Response Effort

Several participants in the AI-assisted condition reported feeling fatigued by the volume of questions. This raises a concern about response quality over the course of the survey: if participants become cognitively drained, later questions may receive less considered answers than earlier ones, potentially undermining the consistency of the data. This effect was not observed in the conventional survey condition, where the fixed format placed a lighter burden on participants.

At the same time, the adaptive mechanism appeared effective in establishing what could be described as a minimum elaboration floor. Even participants who initially provided short or surface-level responses were typically prompted into offering at least some additional clarification, contextualization, or reflection. This suggests that AI-assisted surveys may help mitigate one of the common limitations of static open-ended questionnaires, namely the prevalence of minimal-effort responses.

However, the quality and usefulness of the resulting elaboration varied considerably. In some cases, additional probing generated richer contextual detail, while in others it produced repetitive or low-information responses that appeared driven more by the obligation to continue answering than by genuine participant reflection. This highlights a central limitation of current AI-assisted survey systems: although they can reliably encourage further response production, they remain comparatively limited in their ability to judge when additional probing is unnecessary, counterproductive, or potentially fatiguing for participants. Balancing elaborative depth with participant fatigue therefore comes as an important design consideration for future AI-assisted survey methodologies.

5 Discussion

This study examined whether AI-assisted adaptive surveys can produce richer and more detailed responses than conventional fixed surveys in HCI research. The findings suggest that AI-generated follow-up questions can encourage participants to provide more concrete examples, clarifications, and reflections. However, the results also show that AI-assisted surveys introduce new methodological challenges, such as uneven follow-up quality and increased respondent effort.

However, it is also unable to replicate the contextual understanding and flexibility of a human interviewer. The additional questions posed by the AI survey prompted the respondents to offer concrete examples and clarifications. However, the results also indicate that the AI-assisted survey is unable to understand the underlying meanings of the answers provided by the respondents and is unable to pick up on the cues provided in the answers. The results of the survey indicate a major limitation of the AI-assisted survey. In essence, the results indicate that the survey is most applicable in the exploratory phase of the survey.

When filling out the AI assisted surveys many of the participants expressed that they felt drained by the multiple questions being asked. Participants in the AI-assisted condition reported that the repeated follow-up questions made the survey feel more demanding. This suggests a trade-off between richer qualitative data and increased respondent effort. As participants became fatigued, later responses may have been less detailed, which raises concerns about consistency across the survey.

5.1 Limitations

The study is limited by its small sample size and between-subjects design, which constrain generalizability. The use of a low-stakes topic may not fully reflect the complexity of domain-specific HCI research contexts. Furthermore, the absence of a semi-structured interview condition limits the study's ability to compare AI-assisted surveys with human-led adaptive questioning. These constraints position the study as an exploratory step towards understanding the methodological potential and limitations of AI-assisted adaptive surveys.

Additionally the LLM model that the research team had access to was limited and thus the results of the study may vary if the model used was to change. As the AI field is rapidly evolving [6] these results may also vary greatly in a few years time.

Also it should be noted that the topic of the survey questions might affect the results of the study. As this study was limited by its size the survey questions were centered around a topic that as many people could fill as possible (hobbies).

6 Conclusion

This paper set out to examine whether AI-assisted adaptive surveys can produce richer and more detailed qualitative responses than conventional fixed surveys, addressing a longstanding limitation of survey-based data collection in HCI research: the inability to ask clarifying follow-up questions in response to individual answers.

To investigate this, we developed a lightweight web-based survey platform that uses a large language model to generate adaptive follow-up questions in real time, and compared its output against a conventional fixed survey across eight participants using the same five open-ended base questions on the topic of hobbies.

The results indicate that AI-generated follow-up questions consistently elicited greater elaboration and more concrete detail from participants than the conventional condition, functioning as a minimum elaboration floor. However, the study also identified meaningful limitations: follow-up quality was uneven, the system could not recognise when further probing was unnecessary, and participants in the AI-assisted condition reported higher cognitive effort, raising concerns about response consistency in later questions.

These findings represent a step toward understanding where AI-assisted surveys can add genuine methodological value and where their limitations require careful design consideration. As LLMs become more capable and accessible, this work offers an early empirical grounding for researchers and practitioners seeking to move beyond static surveys without the resource demands of human-led interviewing.

Acknowledgments

We acknowledge the use of Claude (https://claude.ai/) to support early-stage idea generation, including developing background understanding, identifying search terms, and drafting possible base questions for the conventional survey and AI-assisted survey conditions. All final study design decisions, analysis, and writing were conducted by the authors.

Prompts used with Claude were used to come up with questions that could be used both in the new AI assisted survey platform and the regular survey.

- Prompt 1: We are doing a research paper on how AI could be used in research and we are building a LLM assisted survey platform. Come up with five questions we can use for our research as the base questions for the AI-assisted survey and the conventional fixed survey.
- Response 1: Here are five base questions designed to be academically robust, open-ended, and suitable for all three conditions (AI-adaptive survey, static survey, and semi-structured interview). They are phrased to elicit descriptive detail, reflection, and concrete examples, while remaining neutral and comparable across participants. 1. Experience and Description "Can you describe a hobby or activity that you regularly engage in, and what it typically involves?" Opens broadly but anchors in concrete behavior Works well as a starting point for adaptive probing 2. Motivation and Meaning "What motivates you to engage in this activity, and what does it mean to you personally?" Targets intrinsic/extrinsic motivation Encourages reflection beyond surface-level description 3. Context and Practice "Can you walk me through a recent

occasion when you engaged in this activity, including what you did and how you experienced it?" Prompts episodic recall (key for depth) Naturally supports follow-up questions 4. Challenges and Barriers "What challenges or difficulties, if any, have you encountered when engaging in this activity?" Introduces contrast and critical reflection Helps generate richer, more nuanced data 5. Change and Development Over Time "How has your engagement with this activity changed over time, and what factors have influenced those changes?" Captures temporal dimension Encourages causal reasoning and narrative depth

Additionally CLaude was used to ideate different figures that could be used in the project.

- Prompt 2: what kind of figures could i draw for the results section, based on the pilot? (Additionally a file containing the pilot AI survey responses was included with the prompt)
- Response 2: Based on a single pilot response, quantitative figures will be limited, but here are some options that would still be meaningful: Content/Linguistic Analysis Word count per answer: initial vs. follow-up response — a simple bar chart showing how much the follow-up prompt expanded each answer. This directly visualises the "increased elaboration" claim. Response specificity coding — a table or chart scoring each answer on a scale (e.g. vague → concrete) before and after the follow-up, to illustrate depth gain. Thematic/Qualitative Figures A sample interaction flow diagram — showing one Q&A chain (e.g. Q1 about gyoza soup) as a visual flowchart: Base Question → Initial Answer → AI Follow-up → Elaborated Answer. This illustrates how the adaptive loop works in practice. Coding table — a structured table mapping each follow-up question to a category (e.g. clarification, exemplification, elaboration) to show what type of probing the AI defaulted to. Comparative Framework A comparison table: AI survey vs. semi-structured interview — with rows like flexibility, probing depth, consistency, scalability, rated qualitatively. This is common in HCI papers and ties the pilot to your broader argument. Given it's just a pilot, I'd recommend the interaction flow diagram + the word count bar chart as your two core figures — they're honest about the small sample size while still visually grounding your claims. Would you like me to draft either of those?

Reference: Anthropic. (2026). Claude Sonnet 4.6 (Feb 17 version) [Large language model]. https://claude.ai

References

[1] Hamzeh Aljawawdeh, Hamed Fawareh, Mohammed Al Refai, Fuad Alshraideh, Randa Salman, Sara Abu Dabaa't, Heba Alazzeh, Hazim Al-Shdaifat, and Mohammed Khouj. 2024. Ethical Considerations in AI-Driven HCI Systems in Software Engineering. In *2024 25th International Arab Conference on Information Technology (ACIT)*. 1–6. doi:10.1109/ACIT62805.2024.10876926

[2] Marianne Aubin Le Quéré, Hope Schroeder, Casey Randazzo, Jie Gao, Ziv Epstein, Simon Tangi Perrault, David Mimno, Louise Barkhuus, and Hanlin Li. 2024. LLMs as Research Tools: Applications and Evaluations in HCI Data Work. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) *(CHI EA '24)*. Association for Computing Machinery, New York, NY, USA, Article 479, 7 pages. doi:10.1145/3613905.3636301

[3] Ilker Etikan, Sulaiman Abubakar Musa, and Rukayya Sunusi Alkassim. 2015. Comparison of Convenience Sampling and Purposive Sampling. *American Journal of Theoretical and Applied Statistics* 5, 1 (2015), 1–4. arXiv:https://article.sciencepublishinggroup.com/pdf/10.11648.j.ajtas.20160501.11 doi:10.11648/j.ajtas.20160501.11

[4] Kasper Hornbæk, Per Ola Kristensson, and Antti Oulasvirta. 2025. Survey research. In *Introduction to Human-Computer Interaction*. Oxford University Press. arXiv:https://academic.oup.com/book/0/chapter/529000359/chapter-pdf/64021539/oso-9780192864543-chapter-13.pdf doi:10.1093/oso/9780192864543.003.0013

[5] Kehua Lei, Aidan Ladenburg, Zahra Kais Petiwala, Zili Wang, Dishita Jhawar, Ipsita Bisht, Ansh Kumar, and David T. Lee. 2025. Dynamic Surveys: Using LLMs to Blend Qualitative Depth, Quantitative Structure, and Collaborative Interaction. *Proc. ACM Hum.-Comput. Interact.* 9, 7, Article CSCW405 (Oct. 2025), 26 pages. doi:10.1145/3757586

[6] Michael Muller, Lydia B Chilton, Anna Kantosalo, Charles Patrick Martin, and Greg Walsh. 2022. GenAICHI: Generative AI and HCI. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) *(CHI EA '22)*. Association for Computing Machinery, New York, NY, USA, Article 110, 7 pages. doi:10.1145/3491101.3503719

[7] Marianne Aubin Le Quéré, Hope Schroeder, Casey Randazzo, and Jie Gao. 2025. The State of Large Language Models in HCI Research: Workshop Report. *Interactions* 32, 1 (Jan. 2025), 8–9. doi:10.1145/3705617

[8] Ursa Reja, Katja Manfreda, Valentina Hlebec, and Vasja Vehovar. 2003. Open-ended vs. Close-ended Questions in Web Questionnaires. *Adv Methodol Stats* 19 (01 2003).

[9] Jeff Sauro and James R. Lewis. 2016 - 2016. *Quantifying the user experience : practical statistics for user research* (2nd edition. ed.). Morgan Kaufmann is an imprint of Elsevier, Cambridge, MA.